

CNC-Mill — Standard Operating Procedure

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Audience: Maintenance Tech · Line Manager · Quality Worker
Applies to: Line-A, station 2, CNC-Mill

1. Purpose

The CNC-Mill is the second station of Line-A. It receives a turned shaft blank from the CNC-Lathe and machines keyways, flats, and feature pockets into it before the part moves to the Surface-Grinder. A mill outage cascades quickly: the upstream lathe blocks within one buffer fill, and downstream grinders, deburring, and CMM all starve.

2. Normal operating envelope

- Ideal cycle time: **45 s**
- Max acceptable cycle time: **54 s**
- Expected reject rate: **≤ 3 %**
- Fault probability per cycle: **0.8 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

idle_reason	Meaning on the CNC-Mill	First check
Starved	CNC-Lathe (station 1) is not feeding parts.	Inspect the lathe’s status (Fault, Maintenance, or also Idle).
Blocked	Surface-Grinder (station 3) output buffer is full.	Resolve the grinder’s downstream condition first.

4. Faults & corrective actions

4.1 Spindle-Vibration

- **Symptoms in telemetry:** fault_type = "Spindle-Vibration"; reject

rate trends above 3 % in the hour leading up to the fault.

- **Likely root cause:** spindle imbalance, worn tool holder, or loose drawbar.
- **Corrective action:**
 1. LOTO and isolate spindle drive.
 2. Inspect drawbar tension; re-torque to OEM spec.
 3. Replace the tool holder; rebalance the spindle assembly.
 4. Run a 30-second air-cut diagnostic and record vibration amplitude.
- **Restart criteria:** vibration amplitude < 0.5 mm/s RMS at idle.

4.2 Coolant-Leak

- **Symptoms in telemetry:** fault_type = "Coolant-Leak"; operators may also report a visual puddle around the enclosure.
- **Likely root cause:** failed enclosure seal, cracked hose, or loose flood-coolant fitting.
- **Corrective action:**
 1. LOTO and drain the coolant tank.
 2. Trace the leak; replace the failed seal, hose, or fitting.
 3. Refill coolant to spec concentration (5-8 % emulsion).
 4. Run a 5-minute leak test before resuming.
- **Restart criteria:** no visible leak across a 5-minute idle test.

4.3 Tool-Wear

- **Symptoms in telemetry:** fault_type = "Tool-Wear". May be preceded by rising actual cycle time and increasing reject rate, particularly with surface finish callouts at CMM-Inspection.
- **Likely root cause:** insert end of service life; aggressive feed.
- **Corrective action:**
 1. Index the worn tool out of service.
 2. Install fresh insert; reset tool offset against the probe.
 3. Re-inspect any parts machined in the prior 10 cycles at CMM.
- **Restart criteria:** tool offset within 0.005 mm of master.

5. Preventive maintenance schedule

- **Daily:** check coolant concentration; clear chip evacuation auger.
- **Weekly:** spindle drawbar torque check; air-blow tool changer.
- **Monthly:** way-cover inspection; replace coolant filter.
- **Quarterly:** spindle vibration analysis; ATC mechanical service.

6. References

- Maintenance_Workflow_Overview.pdf
- Safety_Lockout_Tagout_Procedure.pdf
- Quality_Reject_Disposition_Policy.pdf
- Line-A_01_CNC-Lathe_SOP.pdf
- Line-A_03_Surface-Grinder_SOP.pdf